

Christopher Legge

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chrislegge 

MEng Computer and Electronic Engineering student at Loughborough University, with hands-on experience in FPGA development, embedded systems design, and machine learning. Finalist in a national engineering design challenge and top 3% globally in an international algorithmic trading competition.

Employment

- Jun 2025 – **Website Designer, Freelance**
 - Jul 2025 Built a fully functioning multi-page website using HTML, CSS and JavaScript, featuring a responsive navigation bar and clean cross-device layout.
- Sep 2023 – **Hockey Coach, Spencer Lynx Hockey Club**
 - Jul 2025 Delivered weekly coaching sessions for 9–11-year-olds, planning drills and managing group dynamics to maintain engagement and safe participation.
- Mar 2024 – **Activity Instructor, Activ Camps**
 - Nov 2024 Led activities for children aged 4–17, balancing safety requirements with high energy delivery to keep participants motivated and engaged.
- Oct 2018 – **Technical Theatre Technician, Trinity School**
 - Jul 2025 Seven years of backstage production experience across 3 shows per year, including 4 musicals. Led sound design and operation for 5 years, then moved to projections — developing skills in live event management, teamwork under pressure, and technical problem-solving.
- Aug 2024 **Engineering Work Experience, InvestIN, UCL University Campus**
 - Two-week intensive programme networking with professional engineers, exploring economic and environmental impacts across multiple engineering disciplines.

Personal Projects

- Jan 2026 – **Vaultly — Full-Stack Document Vault SaaS, Solo Founder**
 - Present Built and deployed a full-stack SaaS product end-to-end as a solo developer: Python (FastAPI) REST API, Next.js web app, and React Native iOS/Android apps, self-hosted on Docker with a Cloudflare tunnel and automated CI/CD. Implemented an OCR pipeline with custom NER to extract data from scanned documents, a secure auth system with JWT refresh token rotation and per-IP rate limiting, and nightly database backups with push-alert on failure. Live at getvaultly.uk with beta users on TestFlight and Google Play.
- Feb 2026 **ML Model — Virtual Scroll Unwrapping, Competition Project**
 - Developed a PyTorch deep learning pipeline to virtually unwrap fragile ancient scrolls from 3D CT scan data, as part of the international Vesuvius Challenge competition.
- Oct 2025 **IoT Data Collection Terminal, Personal**
 - Arduino-based system with multiple connected sensors and a live-updating display. An ESP32 transmits sensor data to a web dashboard for real-time remote monitoring from any location.
- Jun 2025 **Lightsaber, Personal**
 - Designed and built a fully functional toy lightsaber end-to-end: modelled the handle using professional CAD/workshop tools, assembled an LED blade, soldered all electronics, and programmed Arduino animations for power-on and power-off sequences.
- Apr 2024 **Boolean Logic CPU — Tic-Tac-Toe & Connect 4, Personal**
 - Used Minecraft as a 3D logic sandbox to build a working Boolean logic architecture from first principles, implementing fully playable Tic-Tac-Toe and Connect 4 games.

Competitions

May 2026 **IMC Prosperity 4 – Algorithmic Trading Competition, IMC Trading, Ranked 700th of ~18,800 teams – Top 3% globally**

Competed as a two-person team in a global algorithmic trading competition, designing and refining trading algorithms across multiple rounds against thousands of international participants.

Nov 2025 – **Engineers Without Borders Design Challenge – Grand Finalist, Engineers Without Borders UK, Jun 2026 Grand Finals, 19 June 2026 – Top 36 in the UK**

Selected for the national grand finals from hundreds of entries. Built an XGBoost demand prediction model and CVRP-based routing optimiser to dynamically re-route buses and reduce city-wide waiting times. Programmed a DE1-SoC FPGA board in Verilog for live bus location display. Currently one of 36 finalists competing for the national title.

Education

Sep 2025 – **MEng Computer and Electronic Engineering, Loughborough University**

Jun 2030 Predicted: First Class Honours (Year 1, June 2026).

Sep 2018 – **Secondary School, Trinity School, Croydon**

Jun 2025 A Levels: Computer Science, Design & Technology, Mathematics.

Key Skills

Programming C, C++, C#, Python, Java, JavaScript, HTML, CSS, Verilog, Git/GitHub

ML / Data PyTorch, XGBoost, NumPy, SciPy

Engineering Tools Arduino, DE1-SoC FPGA, SolidWorks, NI Multisim, FEMM, RS DesignSpark

Video Editing DaVinci Resolve, Filmora, Final Cut Pro, Blender (3D VFX)

Professional Apps Microsoft Office Suite (Word, Excel, PowerPoint, Outlook, OneNote)

Interests

- **Mountain Biking** – technical trails and long-distance routes including London to Brighton
- **Close-up Magic** – performance confidence and audience engagement
- **Hiking & Camping** – multi-day expeditions; Bronze, Silver & Gold Duke of Edinburgh Award
- **Scuba Diving** – certified diver with experience across multiple countries
- **Hockey** – active club player; teamwork and strategic thinking
- **Guitar** – self-directed learning and regular practice
- **Logic Puzzles**